Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.13	20.23	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.02	+0.5%
power (W), mean	630	631	+0.2%
power (W), peak	656	666	+1.6%
energy (Wh)	211	213	+0.7%
energy, parked workers still on at idle power (Wh)	211	213	+0.7%
CPU used (cores), mean	0.914	0.941	+3.0%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.039	+3.0%
energy per core-hour (Wh)	689	671	-2.7%
node-hours per core-hour	6.57	6.38	-2.9%
pending pods, pod-minutes	0.800	0.000	-100.0%
pending pods, peak	2.00	0.000	-100.0%
HPA replicas, mean	9.12	9.18	+0.7%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.800	0.000	-100.0%
requests timed	1,040	1,145	+10.1%
time over the response line (% of samples)	0.577	0.000	-100.0%
response time (ms), mean	155	50.42	-67.5%
response time (ms), median	130	43.70	-66.3%
response time (ms), 95th percentile	306	73.18	-76.1%
response time (ms), 99th percentile	449	96.22	-78.6%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2207	7.2701	+0.7%	[-1.9665, +2.0503]	EVEN, no significant difference
kwh_per_core_hour	0.7542	0.7593	+0.7%	[-0.1838, +0.2021]	EVEN, no significant difference
utilisation	0.0375	0.0388	+3.4%	[-0.0093, +0.0127]	EVEN, no significant difference
pending_pod_minutes_per_hour	2.2979	0.0000	-100.0%	[-6.1318, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2979	0.0000	-100.0%	[-4.5958, +0.0000]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.391 0.42 0.444 0.466 0.485 0.502 0.518 0.531 0.542 0.552 0.558 0.564 0.568 0.573 0.577 0.58 0.583 0.585 0.586
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	614	5.00	6.00	0.000	0.000	0.612	0.499

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	619	615	7.00	6.00	0.000	0.000	0.627	0.511
2	6.00	6.00	618	619	7.00	6.00	0.000	0.000	0.595	0.617
3	6.00	6.00	623	622	7.00	6.00	0.500	0.000	0.737	0.711
4	6.00	6.00	634	636	9.00	10.00	0.000	0.000	1.01	1.07
5	6.00	6.00	632	636	9.00	10.00	0.000	0.000	0.979	1.11
6	6.00	6.00	634	644	9.00	10.00	0.000	0.000	1.03	1.25
7	6.00	6.00	650	652	9.75	10.00	0.000	0.000	1.48	1.50
8	6.00	6.00	648	654	10.00	10.00	0.000	0.000	1.42	1.56
9	6.00	6.00	646	656	10.00	10.00	0.000	0.000	1.32	1.58
10	6.00	6.00	647	652	10.00	10.00	0.000	0.000	1.38	1.50
11	6.00	6.00	620	616	10.00	10.00	0.000	0.000	0.633	0.538
12	6.00	6.00	618	617	10.00	10.00	0.000	0.000	0.610	0.577
13	6.00	6.00	624	624	10.00	10.00	0.250	0.000	0.766	0.756
14	6.00	6.00	633	636	10.00	10.00	0.000	0.000	0.997	1.08
15	6.00	6.00	636	637	10.00	10.00	0.000	0.000	1.09	1.08
16	6.00	6.00	635	633	10.00	10.00	0.000	0.000	1.04	1.01
17	6.00	6.00	630	623	10.00	10.00	0.000	0.000	0.930	0.727
18	6.00	6.00	618	618	10.00	10.00	0.000	0.000	0.587	0.587
19	6.00	6.00	617	619	10.00	10.00	0.000	0.000	0.570	0.623

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.